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“quarto ## Strengths

1. Clear research focus

- The research question is explicit and narrow: the association between reading/writing scores and mathematics performance.
- The abstract, introduction, and conclusion are aligned with this question.

2. Logical econometric progression

- The sequence of models (simple → multiple → dummy variable → non-linear) reflects standard econometrics pedagogy.
- Each model extends the previous one in a meaningful way.

3. Use of R and Quarto structure

- Code chunks are organized by section and labeled with appropriate chunk options (`echo`, `warning`, `message`, `fig-cap`, etc.).
- The inclusion of `script/estm.R` centralizes estimation code, which is good for reproducibility and separation of concerns.
- A custom `cword()` function for word count shows integration with course tools.

4. Connection to literature and prior work

- The assignment explicitly links to the previous Gretl assignment and references Wickham (2017), which demonstrates awareness of methodological context.

5. Interpretation of models

- Coefficients are interpreted qualitatively (e.g., positive/negative and significance).
- The idea of comparing models via R-squared is introduced, and conclusions about model choice are made.

Weaknesses

1. Overly generic and thin empirical discussion

- The descriptions of results are very general (e.g., “positive and statistically significant”) without:
 - Reporting actual coefficient estimates, standard errors, t-statistics, or confidence intervals.
 - Quantifying economic/educational significance (e.g., “a 10-point increase in reading is associated with X points in math”).
- No discussion of goodness-of-fit beyond a brief statement about R-squared.

2. Code is largely commented out or incomplete

- Many critical R commands are commented out (`#model1 <- ...`, `#plot(...)`, `#summary(data)`, etc.). If run as-is, the document will not produce the claimed analyses or figures.
- In the Introduction and Appendix, the `{{< include "script/estm.R" >}}` syntax is used inside an R chunk, which mixes Quarto shortcodes with knitr syntax in a way that may not work as intended.
- The dataset loading step is missing in the visible code (`data <- ...`) so the analysis is not reproducible from the document alone.

3. Limited data description and exploration

- The “Data Description and Summary” section only states the number of observations and rough means for two variables.
- There is no:
 - Full summary table or descriptive statistics (min, max, sd).
 - Table of variables with definitions and units.
 - Missing values check or data cleaning discussion.
 - Consideration of the categorical variables (parental education, lunch, test prep, gender) beyond a single dummy for female.

4. Underdeveloped diagnostic and robustness analysis

- No regression diagnostics are reported:
 - Residual plots, heteroskedasticity checks, normality of residuals, influential observations, or multicollinearity (especially between reading and writing).
- The non-linear specification is only briefly mentioned, without graphical evidence (e.g., scatterplot with fitted curve) or more flexible forms (e.g., logs, splines).
- Model comparison is based solely on R-squared; no adjusted R-squared, AIC/BIC, or cross-validation are discussed.

5. Weak treatment of econometric issues

- The assignment does not discuss:
 - Potential omitted variable bias (e.g., socio-economic status, school quality, test prep).
 - Endogeneity concerns or causal interpretation (the results are described as associations, but this caveat could be made more explicit).
 - Interpretation of dummy variable coefficient (e.g., “female” shifts in intercept, holding reading constant).
- Significance is mentioned but the choice of significance level, multiple testing, and practical vs. statistical significance are not elaborated.

6. Superficial presentation of visualizations

- The scatterplot chunk is commented out, so no figure is actually produced.
- The text description of the scatterplot is very brief and does not:
 - Comment on dispersion, potential outliers, or non-linearity.
 - Include any regression line or confidence band.
- No additional plots (histograms, boxplots by gender, etc.) are used to understand the data.

7. Quarto/LaTeX structure and citation issues

- There are formatting irregularities:
 - Multiple colons (:) at line ends that look like artifacts from copying; they may cause rendering issues.
 - Code fences like `{r}` instead of standard Quarto`{r}` (unless there is a specific template, this is likely erroneous).
- The reference to `@wickham2017r` is not supported by a visible bibliography file or proper `bibliography:` field in the YAML; reproducible references may not work.
- The `# References` section is an empty div (`::: {#refs}`) without a linked `.bib` file.

8. Conclusions are not fully justified by presented evidence

- The conclusion restates that reading/writing “are important predictors” but:
 - Does not reference magnitudes, diagnostics, or robustness checks.
 - Does not mention any limitations or potential alternative explanations.
- Practical or policy implications (if any) are not discussed, even in a brief, cautious way.

9. Reproducibility and transparency

- Reliance on `script/estm.R` without showing its core content in the main text makes it hard for a reader to understand or reproduce the analysis.
- The name of the dataset file and its loading code are not shown, so an independent reader cannot rerun the assignment from scratch.

Summary

In its current form, the assignment has a well-defined question, a sensible econometric structure, and a good overall layout in Quarto. However, the empirical content is underdeveloped: key code is commented out, numerical results are not reported, diagnostics are missing, and the treatment of data and econometric issues is quite superficial. Strengthening the implementation (uncommenting and verifying code, adding proper summaries and tables) and deepening the interpretation (more detailed results, diagnostics, and discussion of limitations) would significantly improve the quality of the assignment.